JAMES ALEXANDER HARTLEY

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 5 years of hands-on experience building predictive models and deploying ML systems in production environments. Specialise in computer vision and natural language processing with a track record of delivering solutions that reduced operational costs by up to 40%. Comfortable working across the full ML lifecycle from data exploration through to model monitoring and maintenance. Looking to leverage my Python expertise and AWS knowledge to tackle more complex distributed systems challenges.

TECHNICAL SKILLS

Languages: Python, SQL, Java, Scala

Machine Learning: TensorFlow, PyTorch, Scikit-learn, XGBoost, LSTM, CNNs, NLP

Data Processing: Pandas, NumPy, Apache Spark, PySpark

Cloud Platforms: AWS (SageMaker, EC2, S3, Lambda), Google Cloud

Databases: PostgreSQL, MongoDB, DynamoDB

DevOps & Tools: Docker, Git, Jenkins, Jupyter Notebooks, MLflow

Computer Vision: OpenCV, image classification, object detection

NLP: NLTK, spaCy, transformer models, sentiment analysis

WORK EXPERIENCE

Senior Machine Learning Engineer

DataFlow Systems Ltd

www.dataflowsystems.io

Birmingham, United Kingdom

March 2022 - Present

Designed and deployed a real-time anomaly detection system for financial transactions that identified fraudulent activity with 94% precision, saving the company GBP 2.3M annually in prevented losses. Built and maintained 8 production ML models across fraud detection, customer churn prediction, and recommendation systems serving 50M+ daily API requests. Mentored 3 junior ML engineers on best practices for model validation, feature engineering, and reproducible workflows using MLflow. Optimised model inference latency from 850ms to 120ms by implementing model quantisation and deploying on AWS SageMaker endpoints, improving user experience significantly.

Machine Learning Engineer

Luminous AI Solutions

www.luminous-ai.net

Birmingham, United Kingdom

June 2020 - February 2022

Developed a computer vision pipeline for automated quality control in manufacturing that reduced defect detection time by 65%, processing 500+ images per hour with 97% accuracy using convolutional neural networks. Engineered a document classification system using transformer models that processed unstructured business documents, achieving 96% F1 score and eliminating manual review for 10,000+ documents monthly. Implemented comprehensive unit tests and created CI/CD pipelines using Jenkins and Docker, reducing deployment time from 2 hours to 15 minutes. Collaborated with data engineers to design efficient data pipelines on Apache Spark, processing 2TB of daily training data.

Junior Data Scientist

Barclays Bank PLC

www.barclays.com

Birmingham, United Kingdom

August 2018 - May 2020

Built predictive models for customer credit risk assessment using ensemble methods and gradient boosting techniques, contributing to a GBP 15M improvement in loan portfolio performance. Developed Python scripts for automated ETL processes that ingested data from 12 different internal systems into a centralised data warehouse on PostgreSQL. Created interactive dashboards and conducted exploratory data analysis to support business stakeholder decisions across retail and investment banking divisions. Participated in agile ceremonies and delivered 6 machine learning features within strict regulatory compliance frameworks.

EDUCATION

Bachelor of Science in Computer Science

University of Birmingham

Birmingham, United Kingdom

Graduated: June 2018

Grade: 2:1

GITHUB REPOSITORIES

github.com/jahartley/fraud-detection-ensemble - Production fraud detection system using XGBoost and neural networks with model monitoring dashboard

github.com/jahartley/document-classifier-transformers - Document classification pipeline leveraging BERT transformers with fine-tuning scripts and evaluation metrics

github.com/jahartley/cv-quality-control - Computer vision anomaly detection for manufacturing with real-time inference and confidence scoring

github.com/jahartley/ml-utils-library - Reusable utility library for feature engineering, model validation and hyperparameter optimisation

CERTIFICATIONS

AWS Certified Machine Learning - Specialty, Amazon Web Services, 2021

Deep Learning Specialization, Coursera (Andrew Ng), 2020